Comprehensive Data Dictionary — WariMitra

Document Identifier: WM-DD-013

Classification: Physical & Logical Data Element Specifications

Target Database: Google Cloud Firestore NoSQL Database

Standard: ISO/IEC 11179 Metadata Registry Standard

Target Audience: Database Engineers, QA Data Testers, Technical BAs, Integration Architects

1. Collection: `users`

Path: `/users/{userId}`

Description: User accounts, profile references, custom claims, and FCM tokens.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`userId`	`String`	No	Firebase Auth UID (alphanumeric)	Auth UID	Primary identifier matching authentication record.
`email`	`String`	Yes	Valid email format (RFC 5322)	None	Administrative email address.
`phoneNumber`	`String`	Yes	E.164 format (e.g., `+919876543210`)	None	Mobile phone number used for SMS OTP authentication.
`firstName`	`String`	No	Length: 1–50 characters	`""`	User's given first name.
`lastName`	`String`	Yes	Length: 0–50 characters	`""`	User's surname/ family name.
`userType`	`String`	No	Enum: `pilgrim`,	`'pilgrim'`	Role-based

			Enum: `pilgrim`, `volunteer`, `dindi_leader`, `medical_officer`, `police_officer`, `ngo`, `admin`		Role-based authorization claim determining app permissions.
`fcmToken`	`String`	Yes	Valid Firebase Cloud Messaging token	`null`	Device push notification token for emergency broadcast.
`createdAt`	`Timestamp`	No	Server-generated timestamp	`serverTimestamp()`	Account creation date and time.

2. Collection: `sosAlerts`

Path: `/sosAlerts/{alertId}`

Description: Emergency incident logs and spatial coordinates.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`alertId`	`String`	No	Auto-generated document ID	Auto ID	Unique system identifier for the emergency ticket.
`userId`	`String`	No	Valid `users.userId` foreign reference	None	User account who triggered the panic alert.
`status`	`String`	No	Enum: `pending`,	`'pending'`	Current

			Enum: `pending`,		Current

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
			`acknowledged`, `dispatched`, `resolved`		operational lifecycle stage of the incident.
`latitude`	`Float`	No	Range: 8.0 to 35.0 (India bounds)	None	GPS latitude coordinate captured at button trigger.
`longitude`	`Float`	No	Range: 68.0 to 97.5 (India bounds)	None	GPS longitude coordinate captured at button trigger.
`geohash`	`String`	No	9-character Geohash string	None	Spatial index hash used for proximity bounding queries.
`severity`	`String`	No	Enum: `low`, `medium`, `high`, `critical`	`'medium'`	Clinical/tactical urgency level of the emergency.
`description`	`String`	Yes	Max length: 500 characters	`""`	Optional situational or medical symptom notes.
`idempotencyKey`	`String`	No	UUIDv4 (36 characters)	None	Unique client-generated key preventing duplicate writes.
`extraDetails`	`Map`	Yes	JSON key-value	`{}`	Extensible

			JSON key-value		Extensible

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
			map		metadata (blood group, battery percentage).
`createdAtClient`	`Timestamp`	No	Client-supplied timestamp	`Timestamp.now()`	Device clock time when SOS gesture completed.
`createdAt`	`Timestamp`	No	Server-generated timestamp	`serverTimestamp()`	Official server ingestion timestamp for SLA audit.

3. Collection: `medicalCamps`

Path: `/medicalCamps/{campId}`

Description: Physical medical relief posts, clinics, and trauma tents.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`campId`	`String`	No	Auto-generated document ID	Auto ID	Unique identifier of the medical camp facility.
`name`	`String`	No	Length: 3–100 characters	None	Display name of the medical clinic or relief tent.
`latitude`	`Float`	No	Valid geographic latitude	None	Precise physical latitude of the clinic entrance.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`longitude`	`Float`	No	Valid geographic longitude	None	Precise physical longitude of the clinic entrance.
`geohash`	`String`	No	9-character Geohash string	None	Spatial index hash for `getMedicalCamps` query.
`capacity`	`Integer`	No	Value ≥ 0	`0`	Total bed capacity of the medical camp.
`currentPatients`	`Integer`	No	Value ≥ 0	`0`	Number of beds currently occupied by patients.
`staffCount`	`Integer`	No	Value ≥ 0	`1`	Number of doctors and paramedics currently on duty.
`status`	`String`	No	Enum: `active`, `inactive`, `relocating`	`'active'`	Facility availability status.

4. Collection: `familyGroups`

Path: `/familyGroups/{groupId}`

Description: Ad-hoc tracking circles formed by traveling pilgrim groups.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`groupId`	`String`	No	Auto-generated	Auto ID	Unique system

			Auto-generated		Unique system

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
			document ID		identifier for the family circle.
`name`	`String`	No	Length: 2–50 characters	None	Group name chosen by the creator (e.g., "Patil Parivar").
`ownerId`	`String`	No	Valid `users.userId` reference	None	User account who created the circle and holds admin rights.
`inviteCode`	`String`	No	Exact 6-character alphanumeric string	None	Unique joining token shared with family members.
`memberIds`	`Array<String>`	No	Array of valid user IDs	`[ownerId]`	List of all users currently active in the circle.
`createdAt`	`Timestamp`	No	Server-generated timestamp	`serverTimestamp()`	Timestamp when circle was established.

5. Collection: `missingPersons`

Path: `/missingPersons/{reportId}`

Description: Official missing, separated, or lost person intake records.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`reportId`	`String`	No	Auto-generated document ID	Auto ID	Unique case number for the missing person record.
`reportedBy`	`String`	No	Valid `users.userId` reference	None	User ID of the relative or officer filing the report.
`name`	`String`	No	Length: 2–60 characters	None	Full legal name of the missing individual.
`age`	`Integer`	No	Range: 1 to 120	None	Age in years of the missing individual.
`category`	`String`	No	Enum: `child`, `elderly`, `adult`, `differently_abled`	None	Vulnerability classification category.
`photoUrl`	`String`	Yes	Valid HTTPS URL pointing to Cloud Storage	`null`	URL of the photograph uploaded during intake.
`lastSeenLatitude`	`Float`	Yes	Valid latitude coordinates	`null`	Latitude where the person was last spotted.
`lastSeenLongitude`	`Float`	Yes	Valid longitude coordinates	`null`	Longitude where the person was last spotted.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`description`	`String`	No	Max length: 1,000 characters	None	Clothing, distinctive physical traits, or medical conditions.
`status`	`String`	No	Enum: `missing`, `investigating`, `found`, `closed`	`'missing'`	Case investigation status.
`createdAt`	`Timestamp`	No	Server-generated timestamp	`serverTimestamp()`	Date and time when report was lodged.

6. Collection: `blockchainProofs`

Path: `/blockchainProofs/{proofId}`

Description: Cryptographic proof documents anchoring incident history.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`proofId`	`String`	No	Auto-generated document ID	Auto ID	Unique system identifier for the verification proof.
`entityId`	`String`	No	Foreign ID (e.g., `sosAlerts.alertId`)	None	Identifier of the operational record being certified.
`eventType`	`String`	No	Enum:	None	Event name

			Enum:		Event name

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
			`SOS_CREATED`, `SOS_ACKNOWLEDGED`, `SOS_DISPATCHED`, `SOS_RESOLVED`		corresponding to the state transition.
`eventVersion`	`Integer`	No	Integer ≥ 1	`1`	Sequential version counter of proofs for this entity.
`canonicalHash`	`String`	No	64-character hex string (SHA-256)	None	Cryptographic digest of the canonical event snapshot.
`transactionHash`	`String`	Yes	66-character hex string (`0x...`) or `null`	`null`	Public Ethereum Sepolia transaction hash.
`network`	`String`	No	Enum: `sepolia`, `none`	`'none'`	Public blockchain ledger where proof was anchored.
`anchoringStatus`	`String`	No	Enum: `CONFIRMED`, `PENDING`, `LOCAL_PROOF_ONLY`	None	Status of the blockchain write operation.
`snapshot`	`Map`	No	Canonical JSON representation of state	None	Complete frozen payload used to verify hash integrity.

Field Name	Data Type	Nullable?	Validation / Constraints	Default Value	Business Definition & Description
`createdAt`	`Timestamp`	No	Server-generated timestamp	`serverTimestamp()`	Timestamp when proof record was persisted.